SECURITY INCIDENT ANALYSIS REPORT

Date: March 2, 2026

Analyst: Kelwyn Williams

Classification: Lab / Training Exercise

Environment: VMware Virtualized Home Lab

Severity: High — Simulated Threats

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1. Executive Summary

A Wazuh SIEM was deployed in a VMware Workstation virtualized environment consisting of an Ubuntu Server 22.04, Wazuh Manager and a Windows 10 Pro monitored endpoint. Five attack techniques were simulated across three MITRE ATT&CK tactics including Credential Access, Defense Evasion, and Persistence. All five techniques were successfully detected by the Wazuh SIEM and generated actionable alerts with corresponding MITRE ATT&CK mappings. Findings were investigated and documented following standard SOC analyst procedures.

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2. Environment Overview

Component	Details
Hypervisor	VMware Workstation Pro 17
SIEM Platform	Wazuh 4.7
Manager OS	Ubuntu Server 22.04 LTS
Endpoint OS	Windows 10 Pro
Network Configuration	Bridged Adapter
Wazuh Agent Version	4.7

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3. Attack Simulations & Detections

Incident 1 — Brute Force Login Attempts

- **Description:** Multiple consecutive failed login attempts were made against the local Administrator account using incorrect credentials to simulate a brute force attack.
- **Method:** Repeated failed authentication attempts targeting localhost
- **MITRE ATT&CK Tactic:** Credential Access
- **MITRE ATT&CK Technique:** T1110 — Brute Force
- **Windows Event ID:** 4625 — An account failed to log on

- **Wazuh Rule Triggered:** 60204
- **Severity Level:** High
- **Screenshot:**

Mar 2, 2026 @ 13:30:21.986	001	DESKTOP-1C4Q512	T1110	Credential Access	Multiple Windows logon failures.	10	60204

rule.description	Multiple Windows logon failures.
rule.firedtimes	1
rule.frequency	8
rule.gdpr	IV_32.2, IV_35.7.d
rule.groups	windows, windows_security, authentication_failures
rule.hipaa	164.312.b
rule.id	60204
rule.level	10
rule.mail	false
rule.mitre.id	T1110
rule.mitre.tactic	Credential Access
rule.mitre.technique	Brute Force
rule.nist_800_53	AC.7, AU.14, SI.4
rule.pci_dss	10.2.4, 10.2.5, 11.4
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	2026-03-02T18:30:21.986+0000

ID	Name	Created Time	Modified Time	Version
T1110	Brute Force	May 31, 2017 @ 17:31:22.767	Apr 19, 2022 @ 17:28:49.481	2.4

Description

Adversaries may use brute force techniques to gain access to accounts when passwords are unknown or when password hashes are obtained. Without knowledge of the password for an account or set of accounts, an adversary may systematically guess the password using a repetitive or iterative mechanism. Brute forcing passwords can take place via interaction with a service that will check the validity of those credentials or offline against previously acquired credential data, such as password hashes.

Brute forcing credentials may take place at various points during a breach. For example, adversaries may attempt to brute force access to Valid Accounts within a victim environment leveraging knowledge gathered from other post-compromise behaviors such as OS Credential Dumping, Account Discovery, or Password Policy Discovery. Adversaries may also combine brute forcing activity with behaviors such as External Remote Services as part of Initial Access.

Groups

Mitigations

ID	Name	Description
M1018	User Account Management	Manage the creation, modification, use, and permissions associated to user accounts.
M1027	Password Policies	Set and enforce secure password policies for accounts.
M1032	Multi-factor Authentication	Use two or more pieces of evidence to authenticate to a system; such as username and password in addition to a token from a physical smart card or token generator.
T1110	Brute Force Mitigation	Set account lockout policies after a certain number of failed login attempts to prevent passwords from being guessed. Too strict a policy can create a denial of service condition and render environments un-usable, with all accounts being locked-out permanently. Use multifactor authentication. Follow best practices for mitigating access to Valid Accounts Refer to NIST guidelines when creating passwords.(Citation: NIST 800-63-3) Where possible, also enable multi factor authentication on external facing services.
M1036	Account Use Policies	Configure features related to account use like login attempt lockouts, specific login times, etc.

Rows per page: 5 < 1 >

Software

Tactics

ID	Name	Description
TA0006	Credential Access	The adversary is trying to steal account names and passwords. Credential Access consists of techniques for stealing credentials like account names and passwords. Techniques used to get credentials include keylogging or credential dumping. Using legitimate credentials can give adversaries access to systems, make them harder to detect, and provide the opportunity to create more accounts to help achieve their goals.

- **Analyst Notes:** Repeated failed authentications from a single source within a short timeframe is a strong indicator of a brute force attempt. In a real environment this pattern would warrant immediate account lockout and source IP blocking.
- **Recommended Response:** Enforce account lockout policy after 5 failed attempts. Block source IP at firewall. Verify no successful logins occurred before or after the attempts. Alert the affected user.

Incident 2 — File Integrity Violation

- **Description:** A suspicious file was created inside C:\Windows\System32, a protected system directory that should never contain user-generated files.
- **Method:** File created and deleted inside System32 directory
- **MITRE ATT&CK Tactic:** Defense Evasion
- **MITRE ATT&CK Technique:** T1070.004 — Indicator Removal: File Deletion
- **Wazuh Rule Triggered:** 554
- **Severity Level:** High
- **Screenshot:**

@timestamp	2026-03-02T20:48:59.015Z
_id	4FNPsJwBpXEvEpHyul5D
agent.id	001
agent.ip	10.0.0.235
agent.name	DESKTOP-1C4Q512
decoder.name	syscheck_new_entry
full_log	File 'c:\windows\system32\definitely_not_malware.txt' added Mode: realtime
id	1772484539.4349115
input.type	log
location	syscheck
manager.name	wazuh
rule.description	File added to the system.
rule.firedtimes	121
rule.gdpr	II_5.1.f
rule.gpg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_added, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	554
rule.level	5
rule.mail	false
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.event	added
syscheck.md5_after	af2876f5fb8bdfcc683e50a00fc4f822
syscheck.mode	realtime
syscheck.mtime_after	2026-03-02T20:48:58
syscheck.path	c:\windows\system32\definitely_not_malware.txt
syscheck.sha1_after	10038043e2e23aa0cc4256320d5d4831461566b9
syscheck.sha256_after	be7fc6adc347198cc1ea4f8ad24126507f062de1dbc72a795678c6f5dcb89614
syscheck.size_after	56
syscheck.uid_after	S-1-5-32-544
syscheck.uname_after	Administrators
syscheck.win_perm_after	> { "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC",
timestamp	2026-03-02T20:48:59.015+0000

>	Mar 2, 2026 @ 15:57:26.917	001	DESKTOP-1C4Q512	T1070.004 T1485	Defense Evasion, Impact	File deleted.	7	553

@timestamp	2026-03-02T20:57:26.917Z
_id	MVNXsJwBpXEvEpHygl-E
agent.id	001
agent.ip	10.0.0.235
agent.name	DESKTOP-1C4Q512
decoder.name	syscheck_deleted
full_log	File 'c:\windows\system32\definitely_not_malware.txt' deleted Mode: realtime
id	1772485046.4521601
input.type	log
location	syscheck
manager.name	wazuh
rule.description	File deleted.
rule.firedtimes	3
rule.gdpr	II_5.1.f
rule.gpg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_deleted, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	553
rule.level	7
rule.mail	false
rule.mitre.id	T1070.004, T1485
rule.mitre.tactic	Defense Evasion, Impact
rule.mitre.technique	File Deletion, Data Destruction
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PI1.4, PI1.5, CC6.1, CC6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.event	deleted
syscheck.md5_after	af2876f5fb8bdfcc683e50a00fc4f822
syscheck.mode	realtime
syscheck.mtime_after	2026-03-02T20:48:58
syscheck.path	c:\windows\system32\definitely_not_malware.txt
syscheck.sha1_after	10038043e2e23aa0cc4256320d5d4831461566b9
syscheck.sha256_after	be7fc6adc347198cc1ea4f8ad24126507f062de1dbc72a795678c6f5dcb89614
syscheck.size_after	56
syscheck.uid_after	S-1-5-32-544
syscheck.uname_after	Administrators
syscheck.win_perm_after	{ "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC",
timestamp	2026-03-02T20:57:26.917+0000

ID	Name	Created Time	Modified Time	Version
T1070.004	File Deletion	Jan 31, 2020 @ 07:35:36.479	Apr 16, 2022 @ 14:25:43.231	1.1

Description

Adversaries may delete files left behind by the actions of their intrusion activity. Malware, tools, or other non-native files dropped or created on a system by an adversary (ex: Ingress Tool Transfer) may leave traces to indicate to what was done within a network and how. Removal of these files can occur during an intrusion, or as part of a post-intrusion process to minimize the adversary's footprint.

There are tools available from the host operating system to perform cleanup, but adversaries may use other tools as well.(Citation: Microsoft SDelete July 2016) Examples of built-in Command and Scripting Interpreter functions include `del` on Windows and `rm` or `unlink` on Linux and macOS.

Groups

Mitigations

ID	Name	Description
T1107	File Deletion Mitigation	Identify unnecessary system utilities, third-party tools, or potentially malicious software that may be used to delete files, and audit and/or block them by using whitelisting (Citation: Beechey 2010) tools like AppLocker (Citation: Windows Commands JPCERT) (Citation: NSA MS AppLocker) or Software Restriction Policies (Citation: Corio 2008) where appropriate. (Citation: TechNet Applocker vs SRP)

Rows per page: 5 < 1 >

Software

Tactics

ID	Name	Description
TA0005	Defense Evasion	The adversary is trying to avoid being detected. Defense Evasion consists of techniques that adversaries use to avoid detection throughout their compromise. Techniques used for defense evasion include uninstalling/disabling security software or obfuscating/encrypting data and scripts. Adversaries also leverage and abuse trusted processes to hide and masquerade their malware. Other tactics' techniques are cross-listed here when those techniques include the added benefit of subverting defenses.

Rows per page: 5 < 1 >

- **Analyst Notes:** Attackers commonly drop malicious files into System32 because it blends in with legitimate system files. The subsequent deletion indicates an attempt to remove indicators of compromise. File hash was captured by Wazuh FIM at time of creation.
- **Recommended Response:** Investigate the origin of the file creation. Review process creation logs around the same timestamp to identify what process created the file. Escalate to Tier 2 if source cannot be determined.

Incident 3 — New Local Administrator Account

- **Description:** A new local user account was created and immediately added to the local Administrators group, simulating attacker persistence via backdoor account creation.
- **Method:** net user and net localgroup commands executed via PowerShell
- **MITRE ATT&CK Tactic:** Persistence
- **MITRE ATT&CK Technique:** T1098 — Account Manipulation, T1484 – Domain Policy Modification
- **Windows Event ID:** 4720 — New user account created, 4732 — User added to local group
- **Wazuh Rule Triggered:** 60109 / 60154
- **Severity Level:** High
- **Screenshot:**

NEW USER ACCOUNT CREATION

Mar 2, 2026 @ 12:10:23.636	001	DESKTOP-1C4Q512	T1098	Persistence	User account enabled or created.	8	60109

data.win.system.message

```
"A user account was created.

Subject:
    Security ID:        S-1-5-21-3326219252-2042417236-1044767877-1001
    Account Name:       SOCUser
    Account Domain:     DESKTOP-1C4Q512
    Logon ID:           0x272E9D

New Account:
    Security ID:        S-1-5-21-3326219252-2042417236-1044767877-1002
    Account Name:       backdoor
    Account Domain:     DESKTOP-1C4Q512

Attributes:
    SAM Account Name:   backdoor
    Display Name:
    User Principal Name:   -
    Home Directory:
    Home Drive:
    Script Path:
    Profile Path:
    User Workstations:
    Password Last Set:
    Account Expires:
    Primary Group ID:  513
    Allowed To Delegate To:    -
    Old UAC Value:      0x0
    New UAC Value:      0x15
    User Account Control:
        Account Disabled
        'Password Not Required' - Enabled
        'Normal Account' - Enabled
    User Parameters:
    SID History:        -
    Logon Hours:        All

Additional Information:
    Privileges          -"
```

id	1772471423.2242636
input.type	log
location	EventChannel
manager.name	wazuh
rule.description	User account enabled or created.
rule.firedtimes	1
rule.gdpr	IV_32.2, IV_35.7.d
rule.gpg13	7.10
rule.groups	windows, windows_security, account_changed, adduser
rule.hipaa	164.312.a.2.I, 164.312.a.2.II, 164.312.b
rule.id	60109
rule.level	8
rule.mail	false
rule.mitre.id	T1098
rule.mitre.tactic	Persistence
rule.mitre.technique	Account Manipulation
rule.nist_800_53	AC.2, AC.7, AU.14, IA.4
rule.pci_dss	10.2.5, 8.1.2
rule.tsc	CC6.8, CC7.2, CC7.3
timestamp	2026-03-02T17:10:23.636+0000

ID	Name	Created Time	Modified Time	Version
T1098	Account Manipulation	May 31, 2017 @ 17:31:12.196	Apr 18, 2022 @ 10:17:07.164	2.3

Description

Adversaries may manipulate accounts to maintain access to victim systems. Account manipulation may consist of any action that preserves adversary access to a compromised account, such as modifying credentials or permission groups. These actions could also include account activity designed to subvert security policies, such as performing iterative password updates to bypass password duration policies and preserve the life of compromised credentials.

In order to create or manipulate accounts, the adversary must already have sufficient permissions on systems or the domain. However, account manipulation may also lead to privilege escalation where modifications grant access to additional roles, permissions, or higher-privileged Valid Accounts.

Groups

Mitigations

ID	Name	Description
M1026	Privileged Account Management	Manage the creation, modification, use, and permissions associated to privileged accounts, including SYSTEM and root.
M1028	Operating System Configuration	Make configuration changes related to the operating system or a common feature of the operating system that result in system hardening against techniques.
M1030	Network Segmentation	Architect sections of the network to isolate critical systems, functions, or resources. Use physical and logical segmentation to prevent access to potentially sensitive systems and information. Use a DMZ to contain any internet-facing services that should not be exposed from the internal network. Configure separate virtual private cloud (VPC) instances to isolate critical cloud systems.
M1032	Multi-factor Authentication	Use two or more pieces of evidence to authenticate to a system; such as username and password in addition to a token from a physical smart card or token generator.

Rows per page: 5 ‹ 1 ›

Software

Tactics

ID	Name	Description
TA0003	Persistence	The adversary is trying to maintain their foothold. Persistence consists of techniques that adversaries use to keep access to systems across restarts, changed credentials, and other interruptions that could cut off their access. Techniques used for persistence include any access, action, or configuration changes that let them maintain their foothold on systems, such as replacing or hijacking legitimate code or adding startup code.

Rows per page: 5 ‹ 1 ›

USER ACCOUNT ADDED TO ADMIN GROUP

Mar 2, 2026 @ 12:11:02.150	001	DESKTOP-1C4Q512	T1484	Defense Evasion, Privilege Escalation	Administrators group changed.	12	60154

data.win.system.message	"A member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-3326219252-2042417236-1044767877-1001 Account Name: SOCUser Account Domain: DESKTOP-1C4Q512 Logon ID: 0x272E9D Member: Security ID: S-1-5-21-3326219252-2042417236-1044767877-1002 Account Name: - Group: Security ID: S-1-5-32-544 Group Name: Administrators Group Domain: Builtin Additional Information: Privileges: -"
id	1772471462.2258502
input.type	log
location	EventChannel
manager.name	wazuh
rule.description	Administrators group changed.
rule.firedtimes	1
rule.gdpr	IV_32.2, IV_35.7.d
rule.gpg13	7.10
rule.groups	windows, windows_security, group_changed, win_group_changed
rule.hipaa	164.312.a.2.I, 164.312.a.2.II, 164.312.b
rule.id	60154
rule.level	12
rule.mail	true
rule.mitre.id	T1484
rule.mitre.tactic	Defense Evasion, Privilege Escalation
rule.mitre.technique	Domain Policy Modification
rule.nist_800_53	AC.2, AC.7, AU.14, IA.4
rule.pci_dss	10.2.5, 8.1.2
rule.tsc	CC6.8, CC7.2, CC7.3
timestamp	2026-03-02T17:11:02.150+0000

ID	Name	Created Time	Modified Time	Version
T1484	Domain Policy Modification	Mar 7, 2019 @ 09:10:32.650	Feb 9, 2021 @ 10:52:24.598	2.0

Description

Adversaries may modify the configuration settings of a domain to evade defenses and/or escalate privileges in domain environments. Domains provide a centralized means of managing how computer resources (ex: computers, user accounts) can act, and interact with each other, on a network. The policy of the domain also includes configuration settings that may apply between domains in a multi-domain/forest environment. Modifications to domain settings may include altering domain Group Policy Objects (GPOs) or changing trust settings for domains, including federation trusts.

With sufficient permissions, adversaries can modify domain policy settings. Since domain configuration settings control many of the interactions within the Active Directory (AD) environment, there are a great number of potential attacks that can stem from this abuse. Examples of such abuse include modifying GPOs to push a malicious Scheduled Task to computers throughout the domain environment(Citation: ADSecurity GPO Persistence 2016)(Citation: Wald0 Guide to GPOs)(Citation: Harmj0y Abusing GPO Permissions) or modifying domain trusts to include an adversary controlled domain where they can control access tokens that will subsequently be accepted by victim domain resources.(Citation: Microsoft - Customer Guidance on Recent Nation-State Cyber Attacks) Adversaries can also change configuration settings within the AD environment to implement a Rogue Domain Controller.

Adversaries may temporarily modify domain policy, carry out a malicious action(s), and then revert the change to remove suspicious indicators.

> Groups

Mitigations

ID	Name	Description
M1018	User Account Management	Manage the creation, modification, use, and permissions associated to user accounts.
M1026	Privileged Account Management	Manage the creation, modification, use, and permissions associated to privileged accounts, including SYSTEM and root.
M1047	Audit	Perform audits or scans of systems, permissions, insecure software, insecure configurations, etc. to identify potential weaknesses.

Rows per page: 5   1

> Software

Tactics

ID	Name	Description
TA0004	Privilege Escalation	The adversary is trying to gain higher-level permissions. Privilege Escalation consists of techniques that adversaries use to gain higher-level permissions on a system or network. Adversaries can often enter and explore a network with unprivileged access but require elevated permissions to follow through on their objectives. Common approaches are to take advantage of system weaknesses, misconfigurations, and vulnerabilities. Examples of elevated access include: • SYSTEM/root level • local administrator • user account with admin-like access • user accounts with access to specific system or perform specific function These techniques often overlap with Persistence techniques, as OS features that let an adversary persist can execute in an elevated context.
TA0005	Defense Evasion	The adversary is trying to avoid being detected. Defense Evasion consists of techniques that adversaries use to avoid detection throughout their compromise. Techniques used for defense evasion include uninstalling/disabling security software or obfuscating/encrypting data and scripts. Adversaries also leverage and abuse trusted processes to hide and masquerade their malware. Other tactics' techniques are cross-listed here when those techniques include the added benefit of subverting defenses.

Rows per page: 5   1

- **Analyst Notes:** Creation of a new local admin account is one of the most common post-exploitation persistence techniques. The combination of Event ID 4720 followed immediately by 4732 is a strong indicator of malicious intent rather than legitimate IT activity.
- **Recommended Response:** Immediately disable the newly created account. Investigate what process or user created it. Review all admin group membership changes in the past 30 days. Escalate to Tier 2 immediately.

Incident 4 — Windows Event Log Cleared

- **Description:** The System Windows Event Logs were cleared, simulating an attacker attempting to destroy forensic evidence after gaining access to a system.
- **Method:** wevtutil cl System command
- **MITRE ATT&CK Tactic:** Defense Evasion
- **MITRE ATT&CK Technique:** T1070 — Indicator Removal on Host
- **Windows Event ID:** 104 — System log cleared
- **Wazuh Rule Triggered:** 63104
- **Severity Level:** High
- **Screenshot:**

Mar 2, 2026 @ 17:17:15.689	001	DESKTOP-1C4Q512	T1070	Defense Evasion	A Windows log file was cleared	5	63104

@timestamp	2026-03-02T22:17:15.689Z
_id	_1OgsJwBpXEvEpHykmJm
agent.id	001
agent.ip	10.0.0.235
agent.name	DESKTOP-1C4Q512
data.win.logFileCleared.channel	System
data.win.logFileCleared.clientProcessId	7572
data.win.logFileCleared.clientProcessStartKey	281474976712437
data.win.logFileCleared.subjectDomainName	DESKTOP-1C4Q512
data.win.logFileCleared.subjectUserName	SOCUser
data.win.system.channel	System
data.win.system.computer	DESKTOP-1C4Q512
data.win.system.eventID	104
data.win.system.eventRecordID	669
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"The System log file was cleared."
data.win.system.opcode	0
data.win.system.processID	1424
data.win.system.providerGuid	{fc65ddd8-d6ef-4962-83d5-6e5cfe9ce148}
data.win.system.providerName	Microsoft-Windows-Eventlog
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-03-02T22:17:15.4097448Z
data.win.system.task	104
data.win.system.threadID	5960
data.win.system.version	1
decoder.name	windows_eventchannel
id	1772489835.6919634
input.type	log
location	EventChannel
manager.name	wazuh
rule.description	A Windows log file was cleared
rule.firedtimes	1
rule.gdpr	II_5.1.f
rule.gpg13	10.1
rule.groups	windows, windows_logs, log_clearing
rule.id	63104
rule.level	5
rule.mail	false
rule.mitre.id	T1070
rule.mitre.tactic	Defense Evasion
rule.mitre.technique	Indicator Removal on Host
timestamp	2026-03-02T22:17:15.689+0000

ID	Name	Created Time	Modified Time	Version
T1070	Indicator Removal on Host	May 31, 2017 @ 17:30:55.892	Apr 1, 2022 @ 16:24:14.332	1.3

Description

Adversaries may delete or modify artifacts generated on a host system to remove evidence of their presence or hinder defenses. Various artifacts may be created by an adversary or something that can be attributed to an adversary's actions. Typically these artifacts are used as defensive indicators related to monitored events, such as strings from downloaded files, logs that are generated from user actions, and other data analyzed by defenders. Location, format, and type of artifact (such as command or login history) are often specific to each platform.

Removal of these indicators may interfere with event collection, reporting, or other processes used to detect intrusion activity. This may compromise the integrity of security solutions by causing notable events to go unreported. This activity may also impede forensic analysis and incident response, due to lack of sufficient data to determine what occurred.

> Groups

Mitigations

ID	Name	Description
M1022	Restrict File and Directory Permissions	Restrict access by setting directory and file permissions that are not specific to users or privileged accounts.
M1029	Remote Data Storage	Use remote security log and sensitive file storage where access can be controlled better to prevent exposure of intrusion detection log data or sensitive information.
T1070	Indicator Removal on Host Mitigation	Automatically forward events to a log server or data repository to prevent conditions in which the adversary can locate and manipulate data on the local system. When possible, minimize time delay on event reporting to avoid prolonged storage on the local system. Protect generated event files that are stored locally with proper permissions and authentication and limit opportunities for adversaries to increase privileges by preventing Privilege Escalation opportunities. Obfuscate/encrypt event files locally and in transit to avoid giving feedback to an adversary.
M1041	Encrypt Sensitive Information	Protect sensitive information with strong encryption.

Rows per page: 5 ‹ 1 ›

> Software

Tactics

ID	Name	Description
TA0005	Defense Evasion	The adversary is trying to avoid being detected. Defense Evasion consists of techniques that adversaries use to avoid detection throughout their compromise. Techniques used for defense evasion include uninstalling/disabling security software or obfuscating/encrypting data and scripts. Adversaries also leverage and abuse trusted processes to hide and masquerade their malware. Other tactics' techniques are cross-listed here when those techniques include the added benefit of subverting defenses.

Rows per page: 5 ‹ 1 ›

- **Analyst Notes:** Clearing event logs is almost never legitimate administrator behavior and should always be treated as a high priority alert. The timing of this event relative to other detections suggests an attacker attempting to cover their tracks after prior malicious activity.
- **Recommended Response:** Treat as high priority. Immediately preserve all remaining logs. Check if logs were forwarded to SIEM before clearing — Wazuh captures events in real time so cleared local logs do not destroy SIEM records. Escalate to Tier 2 and begin full incident response procedure.

Incident 5 — Malicious Service Creation

- **Description:** A new Windows service was created pointing to cmd.exe, simulating an attacker establishing persistent access that survives system reboots.
- **Method:** sc create command executed via Command Prompt as Administrator
- **MITRE ATT&CK Tactic:** Persistence
- **MITRE ATT&CK Technique:** T1543.003 — Create or Modify System Process: Windows Service
- **Windows Event ID:** 7045 — New Service Installed
- **Wazuh Rule Triggered:** 61138
- **Severity Level:** High
- **Screenshot:**

Mar 2, 2026 @ 17:30:08.005	001	DESKTOP-1C40512	T1543.003	Persistence, Privilege Escalation	New Windows Service Created	5	61138

@timestamp	2026-03-02T22:30:08.005Z
_id	mFOssJwBpXEvEpHyWmM5
agent.id	001
agent.ip	10.0.0.235
agent.name	DESKTOP-1C4Q512
data.win.eventdata.accountName	LocalSystem
data.win.eventdata.imagePath	C:\\Windows\\System32\\cmd.exe
data.win.eventdata.serviceName	FakeService
data.win.eventdata.serviceType	user mode service
data.win.eventdata.startType	auto start
data.win.system.channel	System
data.win.system.computer	DESKTOP-1C4Q512
data.win.system.eventID	7045
data.win.system.eventRecordID	671
data.win.system.eventSourceName	Service Control Manager
data.win.system.keywords	0x8080000000000000
data.win.system.level	4
data.win.system.message	"A service was installed in the system. Service Name: FakeService Service File Name: C:\Windows\System32\cmd.exe Service Type: user mode service Service Start Type: auto start Service Account: LocalSystem"
data.win.system.opcode	0
data.win.system.processID	664
data.win.system.providerGuid	{555908d1-a6d7-4695-8e1e-26931d2012f4}
data.win.system.providerName	Service Control Manager
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-03-02T22:30:06.7397648Z
data.win.system.task	0
data.win.system.threadID	6504
data.win.system.version	0
decoder.name	windows_eventchannel
id	1772490608.7290103
input.type	log
location	EventChannel
manager.name	wazuh
rule.description	New Windows Service Created
rule.firedtimes	1
rule.groups	windows, windows_system
rule.id	61138
rule.level	5
rule.mail	false
rule.mitre.id	T1543.003
rule.mitre.tactic	Persistence, Privilege Escalation
rule.mitre.technique	Windows Service
timestamp	2026-03-02T22:30:08.005+0000

ID	Name	Created Time	Modified Time	Version
T1543.003	Windows Service	Jan 17, 2020 @ 14:13:50.402	May 20, 2022 @ 12:22:32.605	1.2

Description

Adversaries may create or modify Windows services to repeatedly execute malicious payloads as part of persistence. When Windows boots up, it starts programs or applications called services that perform background system functions.(Citation: TechNet Services) Windows service configuration information, including the file path to the service's executable or recovery programs/commands, is stored in the Windows Registry.

Adversaries may install a new service or modify an existing service to execute at startup in order to persist on a system. Service configurations can be set or modified using system utilities (such as sc.exe), by directly modifying the Registry, or by interacting directly with the Windows API.

Adversaries may also use services to install and execute malicious drivers. For example, after dropping a driver file (ex: `.sys`) to disk, the payload can be loaded and registered via Native API functions such as `CreateServiceW()` (or manually via functions such as `ZwLoadDriver()` and `ZwSetValueKey()`), by creating the required service Registry values (i.e. Modify Registry), or by using command-line utilities such as `PsExtl.exe`.(Citation: Symantec W.32 Stuxnet Dossier)(Citation: Crowdstrike DriveSlayer February 2022)(Citation: Unit42 AcidBox June 2020) Adversaries may leverage these drivers as Rootkits to hide the presence of malicious activity on a system. Adversaries may also load a signed yet vulnerable driver onto a compromised machine (known as "Bring Your Own Vulnerable Driver" (BYOVD)) as part of Exploitation for Privilege Escalation.(Citation: ESET InvisiMole June 2020)(Citation: Unit42 AcidBox June 2020)

Services may be created with administrator privileges but are executed under SYSTEM privileges, so an adversary may also use a service to escalate privileges. Adversaries may also directly start services through Service Execution. To make detection analysis more challenging, malicious services may also incorporate Masquerade Task or Service (ex: using a service and/or payload name related to a legitimate OS or benign software component).

> Groups

Mitigations

ID	Name	Description
M1018	User Account Management	Manage the creation, modification, use, and permissions associated to user accounts.
M1028	Operating System Configuration	Make configuration changes related to the operating system or a common feature of the operating system that result in system hardening against techniques.
M1045	Code Signing	Enforce binary and application integrity with digital signature verification to prevent untrusted code from executing.
M1040	Behavior Prevention on Endpoint	Use capabilities to prevent suspicious behavior patterns from occurring on endpoint systems. This could include suspicious process, file, API call, etc. behavior.
M1047	Audit	Perform audits or scans of systems, permissions, insecure software, insecure configurations, etc. to identify potential weaknesses.

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> Software

Tactics

ID	Name	Description
TA0004	Privilege Escalation	The adversary is trying to gain higher-level permissions. Privilege Escalation consists of techniques that adversaries use to gain higher-level permissions on a system or network. Adversaries can often enter and explore a network with unprivileged access but require elevated permissions to follow through on their objectives. Common approaches are to take advantage of system weaknesses, misconfigurations, and vulnerabilities. Examples of elevated access include: • SYSTEM/root level • local administrator • user account with admin-like access • user accounts with access to specific system or perform specific function These techniques often overlap with Persistence techniques, as OS features that let an adversary persist can execute in an elevated context.
TA0003	Persistence	The adversary is trying to maintain their foothold. Persistence consists of techniques that adversaries use to keep access to systems across restarts, changed credentials, and other interruptions that could cut off their access. Techniques used for persistence include any access, action, or configuration changes that let them maintain their foothold on systems, such as replacing or hijacking legitimate code or adding startup code.

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- **Analyst Notes:** Legitimate software rarely creates services pointing directly to cmd.exe. This is a strong indicator of malicious persistence. The service was configured to start automatically ensuring it would survive reboots.
- **Recommended Response:** Immediately disable and delete the suspicious service. Investigate what account created it and when. Review all service creation events in the past 7 days. Determine if the system needs to be reimaged.

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4. Windows Event ID Reference

Event ID	Description	Generated By
4625	Failed login attempt	Brute force simulation
4720	New user account created	Backdoor account creation
4732	User added to local group	Privilege escalation
1102	Audit log was cleared	Event log clearing
104	System log was cleared	Event log clearing
7045	New service installed	Malicious service creation

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5. MITRE ATT&CK Coverage Summary

Tactic	Technique	ID
Credential Access	Brute Force	T1110
Defense Evasion	File Deletion	T1070.004
Persistence	Account Manipulation	T1098
Persistence	Domain Policy Modification	T1484
Defense Evasion	Clear Windows Event Logs	T1070.001
Persistence	Windows Service Creation	T1543.003

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6. Detection Gaps & Tuning Notes

During this exercise several detection configurations required manual tuning before alerts fired correctly. This section documents those findings as they reflect real SOC engineering work.

- **File Integrity Monitoring** required adding System32 to the monitored directories in ossec.conf with realtime="yes" enabled. Default Wazuh configuration does not monitor this directory out of time.
- **PowerShell Script Block Logging** required manual enablement via registry before Wazuh could detect encoded PowerShell execution. This represents a common detection gap in default Windows configurations.
- **Registry Monitoring** required explicit configuration of windows_registry entries in ossec.conf before FIM could detect Run key modifications.

This section demonstrates that out of the box SIEM configurations require tuning to achieve adequate detection coverage — a core responsibility of SOC analysts and detection engineers.

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7. Remediation Recommendations

- Enforce account lockout after 5 consecutive failed login attempts across all endpoints
- Alert on all new local account creation and group membership changes in real time
- Treat any event log clearing as a critical priority alert requiring immediate escalation
- Monitor all new Windows service creation and alert on services pointing to cmd.exe or powershell.exe
- Enable PowerShell Script Block Logging via Group Policy across all endpoints
- Configure Wazuh FIM with realtime monitoring on all sensitive directories including System32, Program Files, and user profile directories
- Forward all Windows Event Logs to SIEM in real time so local log clearing cannot destroy forensic evidence

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8. Conclusion

This lab exercise successfully validated the Wazuh SIEM's capability to detect common Windows-based attack techniques in real time across three MITRE ATT&CK tactics. All five simulated attacks generated corresponding alerts with accurate MITRE ATT&CK mappings demonstrating effective detection coverage. The exercise also highlighted the importance of proper SIEM tuning, as several detections required manual configuration before alerts fired correctly. This reflects real world SOC operations where continuous detection tuning is as important as initial deployment.